CACTUS FALL 2026 · WEEK 6 · LEARNING SESSION

AI as annotator

An AI can label thousands of texts in minutes. Whether you can trust the labels is a measurement question.

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THE IDEA

Fast labels, unknown quality

The promise

A label for every text, in minutes, with the same wording each time.

The risk

It can be confident and wrong, and nothing on the screen tells you which.

The rule

Treat the AI as a measuring instrument. Test it against people before you trust it.

Today: how to write a codebook, how to prompt, how to measure agreement, and how to read the disagreements.

START WITH A CODEBOOK

The AI can only follow what you write

A definition

One sentence. What counts as a yes, and what does not.

Examples

Two or three clear yes, two or three clear no, and the hard ones in between.

Edge cases

What to do when it could go either way. Write the rule down.

People and the AI need the **same text**. If they disagree later, the first thing to check is whether the codebook settled the case.

LIVE DEMO

From codebook to prompt

```
You are labeling social media messages for a research project.
For each numbered message, answer yes or no:
Does the message give a reason for its claim?
A reason is an explanation, evidence, or an example.
Reply with one line per message: the number, then yes or no.
Do not explain.
```

One label per prompt. A fixed answer format. Then **record the model, the prompt, and the date**: the same prompt can give different answers tomorrow.

COMPARE WITH PEOPLE

Four counts, four measures

	AI: yes	AI: no
People: yes	4 hits	2 missed
People: no	4 false alarms	90 correct nos

Accuracy: how often they agree (94 of 100).

Recall: of the true yes, how many it found (4 of 6).

Precision: of its yes, how many are right (4 of 8).

Kappa: agreement beyond chance (next slide).

Invented numbers. Always say how many texts, and which people labeled them.

THE ACCURACY TRAP

Always saying no scores 94%

Labeler	Accuracy	Finds	Flags right
A: always says no	94%	0%	n/a
B: flags a lot	75%	83%	17%
C: balanced	**94%**	**67%**	**50%**

When the thing you look for is rare, accuracy hides what you care about. Compare with the **always-no baseline**, and report recall and precision. (Invented numbers: 6 of 100 messages are truly yes.)

AGREEMENT BEYOND CHANCE

What kappa asks

The problem

Two labelers who both say no most of the time agree often by luck.

The idea

Kappa takes the agreement you got, removes the agreement chance alone would give, and rescales.

Reading it

Near 0: no better than chance. Near 1: almost perfect. Read it with the counts, not alone.

A labeler can have 90% accuracy and a kappa near zero. That is why you report more than one number.

READ THE DISAGREEMENTS

Three reasons they disagree

The labeler is wrong

"Your plan is idiotic." The AI says abusive, but the target is a plan, not a person.

The codebook does not say

"Typical politician, lies every time." A person, or a group? Either reading is fair.

The human label is questionable

"You are a clown" labeled not abusive. A slip, or a different rule?

The numbers say **how often**. Only reading says **why**, and why tells you whether to change the prompt, the codebook, or the human labels.

REPORT IT HONESTLY

What to write down

The setup

Model name and date, the full prompt, and the codebook the people used.

The numbers

How many texts, from where, accuracy and the baseline, recall, precision, kappa, and how much two people agreed.

The limits

Tune on one set, score a separate set once. One source is not every platform. Humans are not ground truth.

Never paste private or identifying data into a chat tool. Use public, de-identified text, or a tool your ethics process allows.

WHAT TO DO NEXT

Before the coding session

Practice page

Can You Trust the Labeler?, about 20 minutes. It ends with the measures on 20 labels.

Required reading

Textbook Chapter 11, sections 11.1 and 11.4, and Chapter 8, section 8.5.

Data

The class messages are on the class drive. They include abusive language. Course use only.

In the coding session: measure agreement on real messages, read the disagreements, run your own labeling, and find three mistakes in AI-written code.

QUESTIONS

What would you check first?

Imagine an AI labeled your project's texts. What is the first number you would ask for, and why?